Grid analysis report

Monday, December 15, 2025, 02:01:42

Station Name	500-220KV Project
Project	500-220KV Project
Study	500-220KV Project

Parameters

Equivalent Parallel Z Spec.	User Defined
Split Factor	0.7
Nominal Frequency	50 hz
Bus ID	500KV
LG Fault Current	30000 amps
Remote Contribution	100 %
Upper Layer Thickness	4.69 meters
Upper Layer Resistivity	168.26 ohm-m
Lower Layer Resistivity	83.02 ohm-m

Output Results

Parallel Z	0.627018 ohms
Decrement Factor	1.01579

Primary Electrode Elements

X1	Y1	Z1	X2	Y2	Z2	Length	Radius	Current	Electrode
(meters)							(mm)	(amps)	(#)
0	0	0.5	75	75	0.5	106.066	8.7	82.7724	Asy 1
75	75	0.5	150	150	0.5	106.066	8.7	79.1413	Asy 1
0	150	0.5	75	75	0.5	106.066	8.7	79.8083	Asy 2
75	75	0.5	150	0	0.5	106.066	8.7	80.5985	Asy 2
0.5	0	0.5	0.5	75	0.5	75	8.7	103.3036	Asy 3
0.5	75	0.5	0.5	150	0.5	75	8.7	103.2973	Asy 3
1	0	0.5	1	75	0.5	75	8.7	74.9641	Asy 4
1	75	0.5	1	150	0.5	75	8.7	74.9401	Asy 4
1.5	0	0.5	1.5	75	0.5	75	8.7	64.9501	Asy 5
1.5	75	0.5	1.5	150	0.5	75	8.7	64.9102	Asy 5
2	0	0.5	2	75	0.5	75	8.7	66.3541	Asy 6
2	75	0.5	2	150	0.5	75	8.7	66.2923	Asy 6
2.5	0	0.5	2.5	75	0.5	75	8.7	87.4255	Asy 7
2.5	75	0.5	2.5	150	0.5	75	8.7	87.3394	Asy 7
149.5	0	0.5	149.5	75	0.5	75	8.7	103.2651	Asy 8
149.5	75	0.5	149.5	150	0.5	75	8.7	103.1676	Asy 8
149	0	0.5	149	75	0.5	75	8.7	74.9129	Asy 9
149	75	0.5	149	150	0.5	75	8.7	74.7937	Asy 9
148.5	0	0.5	148.5	75	0.5	75	8.7	64.8597	Asy 10
148.5	75	0.5	148.5	150	0.5	75	8.7	64.7086	Asy 10
148	0	0.5	148	75	0.5	75	8.7	66.1722	Asy 11
148	75	0.5	148	150	0.5	75	8.7	65.9704	Asy 11
147.5	0	0.5	147.5	75	0.5	75	8.7	87.0018	Asy 12
147.5	75	0.5	147.5	150	0.5	75	8.7	86.7118	Asy 12
0	0.5	0.5	75	0.5	0.5	75	8.7	99.6204	Asy 13
75	0.5	0.5	150	0.5	0.5	75	8.7	99.5784	Asy 13
0	1	0.5	75	1	0.5	75	8.7	72.3225	Asy 14
75	1	0.5	150	1	0.5	75	8.7	72.2786	Asy 14
0	1.5	0.5	75	1.5	0.5	75	8.7	62.7407	Asy 15
75	1.5	0.5	150	1.5	0.5	75	8.7	62.6715	Asy 15
0	2	0.5	75	2	0.5	75	8.7	64.1005	Asy 16
75	2	0.5	150	2	0.5	75	8.7	63.9635	Asy 16

0	2.5	0.5	75	2.5	0.5	75	8.7	84.2993	Asy 17
75	2.5	0.5	150	2.5	0.5	75	8.7	83.9677	Asy 17
0	149.5	0.5	75	149.5	0.5	75	8.7	101.5553	Asy 18
75	149.5	0.5	150	149.5	0.5	75	8.7	101.534	Asy 18
0	149	0.5	75	149	0.5	75	8.7	74.6333	Asy 19
75	149	0.5	150	149	0.5	75	8.7	74.6194	Asy 19
0	148.5	0.5	75	148.5	0.5	75	8.7	65.9378	Asy 20
75	148.5	0.5	150	148.5	0.5	75	8.7	65.9229	Asy 20
0	148	0.5	75	148	0.5	75	8.7	69.1633	Asy 21
75	148	0.5	150	148	0.5	75	8.7	69.1391	Asy 21
0	147.5	0.5	75	147.5	0.5	75	8.7	93.9055	Asy 22
75	147.5	0.5	150	147.5	0.5	75	8.7	93.8502	Asy 22
0	0	0.5	75	0	0.5	75	8.7	180.2473	Sym 1
75	0	0.5	150	0	0.5	75	8.7	180.1716	Sym 1
0	5.1724	0.5	75	5.1724	0.5	75	8.7	128.4543	Sym 1
75	5.1724	0.5	150	5.1724	0.5	75	8.7	126.334	Sym 1
0	10.3448	0.5	75	10.3448	0.5	75	8.7	171.3918	Sym 1
75	10.3448	0.5	150	10.3448	0.5	75	8.7	171.1388	Sym 1
0	15.5172	0.5	75	15.5172	0.5	75	8.7	159.6609	Sym 1
75	15.5172	0.5	150	15.5172	0.5	75	8.7	159.6759	Sym 1
0	20.6897	0.5	75	20.6897	0.5	75	8.7	147.519	Sym 1
75	20.6897	0.5	150	20.6897	0.5	75	8.7	147.5908	Sym 1
0	25.8621	0.5	75	25.8621	0.5	75	8.7	135.478	Sym 1
75	25.8621	0.5	150	25.8621	0.5	75	8.7	135.577	Sym 1
0	31.0345	0.5	75	31.0345	0.5	75	8.7	127.6282	Sym 1
75	31.0345	0.5	150	31.0345	0.5	75	8.7	127.7447	Sym 1
0	36.2069	0.5	75	36.2069	0.5	75	8.7	119.5299	Sym 1
75	36.2069	0.5	150	36.2069	0.5	75	8.7	119.6552	Sym 1
0	41.3793	0.5	75	41.3793	0.5	75	8.7	114.5382	Sym 1
75	41.3793	0.5	150	41.3793	0.5	75	8.7	114.6706	Sym 1
0	46.5517	0.5	75	46.5517	0.5	75	8.7	109.2295	Sym 1
75	46.5517	0.5	150	46.5517	0.5	75	8.7	109.3627	Sym 1
0	51.7241	0.5	75	51.7241	0.5	75	8.7	106.0474	Sym 1
75	51.7241	0.5	150	51.7241	0.5	75	8.7	106.1803	Sym 1
0	56.8966	0.5	75	56.8966	0.5	75	8.7	102.71	Sym 1
75	56.8966	0.5	150	56.8966	0.5	75	8.7	102.8373	Sym 1
0	62.069	0.5	75	62.069	0.5	75	8.7	100.7795	Sym 1
75	62.069	0.5	150	62.069	0.5	75	8.7	100.8987	Sym 1
0	67.2414	0.5	75	67.2414	0.5	75	8.7	99.683	Sym 1
75	67.2414	0.5	150	67.2414	0.5	75	8.7	99.793	Sym 1
0	72.4138	0.5	75	72.4138	0.5	75	8.7	99.5104	Sym 1
75	72.4138	0.5	150	72.4138	0.5	75	8.7	99.6018	Sym 1
0	77.5862	0.5	75	77.5862	0.5	75	8.7	99.9905	Sym 1
75	77.5862	0.5	150	77.5862	0.5	75	8.7	100.0945	Sym 1
0	82.7586	0.5	75	82.7586	0.5	75	8.7	100.5772	Sym 1
75	82.7586	0.5	150	82.7586	0.5	75	8.7	100.6947	Sym 1
0	87.931	0.5	75	87.931	0.5	75	8.7	102.0458	Sym 1
75	87.931	0.5	150	87.931	0.5	75	8.7	102.1678	Sym 1
0	93.1034	0.5	75	93.1034	0.5	75	8.7	103.5587	Sym 1
75	93.1034	0.5	150	93.1034	0.5	75	8.7	103.6788	Sym 1
0	98.2759	0.5	75	98.2759	0.5	75	8.7	106.474	Sym 1
75	98.2759	0.5	150	98.2759	0.5	75	8.7	106.5918	Sym 1
0	103.4483	0.5	75	103.4483	0.5	75	8.7	109.2704	Sym 1
75	103.4483	0.5	150	103.4483	0.5	75	8.7	109.3804	Sym 1
0	108.6207	0.5	75	108.6207	0.5	75	8.7	114.7447	Sym 1
75	108.6207	0.5	150	108.6207	0.5	75	8.7	114.853	Sym 1
0	113.7931	0.5	75	113.7931	0.5	75	8.7	119.8354	Sym 1
75	113.7931	0.5	150	113.7931	0.5	75	8.7	119.9328	Sym 1
0	118.9655	0.5	75	118.9655	0.5	75	8.7	127.8925	Sym 1

75	118.9655	0.5	150	118.9655	0.5	75	8.7	127.9765	Sym 1
0	124.1379	0.5	75	124.1379	0.5	75	8.7	135.4934	Sym 1
75	124.1379	0.5	150	124.1379	0.5	75	8.7	135.5483	Sym 1
0	129.3103	0.5	75	129.3103	0.5	75	8.7	146.682	Sym 1
75	129.3103	0.5	150	129.3103	0.5	75	8.7	146.6718	Sym 1
0	134.4828	0.5	75	134.4828	0.5	75	8.7	153.6402	Sym 1
75	134.4828	0.5	150	134.4828	0.5	75	8.7	153.3751	Sym 1
0	139.6552	0.5	75	139.6552	0.5	75	8.7	141.0206	Sym 1
75	139.6552	0.5	150	139.6552	0.5	75	8.7	138.7883	Sym 1
0	144.8276	0.5	75	144.8276	0.5	75	8.7	155.7692	Sym 1
75	144.8276	0.5	150	144.8276	0.5	75	8.7	155.3708	Sym 1
0	150	0.5	75	150	0.5	75	8.7	182.359	Sym 1
75	150	0.5	150	150	0.5	75	8.7	182.3049	Sym 1
0	0	0.5	0	75	0.5	75	8.7	186.4806	Sym 1
0	75	0.5	0	150	0.5	75	8.7	186.532	Sym 1
5.1724	0	0.5	5.1724	75	0.5	75	8.7	130.8826	Sym 1
5.1724	75	0.5	5.1724	150	0.5	75	8.7	131.3758	Sym 1
10.3448	0	0.5	10.3448	75	0.5	75	8.7	171.3728	Sym 1
10.3448	75	0.5	10.3448	150	0.5	75	8.7	170.9643	Sym 1
15.5172	0	0.5	15.5172	75	0.5	75	8.7	159.8065	Sym 1
15.5172	75	0.5	15.5172	150	0.5	75	8.7	159.1973	Sym 1
20.6897	0	0.5	20.6897	75	0.5	75	8.7	147.6339	Sym 1
20.6897	75	0.5	20.6897	150	0.5	75	8.7	146.9161	Sym 1
25.8621	0	0.5	25.8621	75	0.5	75	8.7	136.1223	Sym 1
25.8621	75	0.5	25.8621	150	0.5	75	8.7	135.3315	Sym 1
31.0345	0	0.5	31.0345	75	0.5	75	8.7	126.3866	Sym 1
31.0345	75	0.5	31.0345	150	0.5	75	8.7	125.5381	Sym 1
36.2069	0	0.5	36.2069	75	0.5	75	8.7	119.9766	Sym 1
36.2069	75	0.5	36.2069	150	0.5	75	8.7	119.1109	Sym 1
41.3793	0	0.5	41.3793	75	0.5	75	8.7	113.8323	Sym 1
41.3793	75	0.5	41.3793	150	0.5	75	8.7	112.9499	Sym 1
46.5517	0	0.5	46.5517	75	0.5	75	8.7	108.8136	Sym 1
46.5517	75	0.5	46.5517	150	0.5	75	8.7	107.9167	Sym 1
51.7241	0	0.5	51.7241	75	0.5	75	8.7	105.9159	Sym 1
51.7241	75	0.5	51.7241	150	0.5	75	8.7	105.0104	Sym 1
56.8966	0	0.5	56.8966	75	0.5	75	8.7	102.7389	Sym 1
56.8966	75	0.5	56.8966	150	0.5	75	8.7	101.8231	Sym 1
62.069	0	0.5	62.069	75	0.5	75	8.7	100.5172	Sym 1
62.069	75	0.5	62.069	150	0.5	75	8.7	99.6008	Sym 1
67.2414	0	0.5	67.2414	75	0.5	75	8.7	99.772	Sym 1
67.2414	75	0.5	67.2414	150	0.5	75	8.7	98.8488	Sym 1
72.4138	0	0.5	72.4138	75	0.5	75	8.7	98.6388	Sym 1
72.4138	75	0.5	72.4138	150	0.5	75	8.7	97.6805	Sym 1
77.5862	0	0.5	77.5862	75	0.5	75	8.7	98.506	Sym 1
77.5862	75	0.5	77.5862	150	0.5	75	8.7	97.5426	Sym 1
82.7586	0	0.5	82.7586	75	0.5	75	8.7	99.7391	Sym 1
82.7586	75	0.5	82.7586	150	0.5	75	8.7	98.7987	Sym 1
87.931	0	0.5	87.931	75	0.5	75	8.7	100.5384	Sym 1
87.931	75	0.5	87.931	150	0.5	75	8.7	99.5923	Sym 1
93.1034	0	0.5	93.1034	75	0.5	75	8.7	102.7934	Sym 1
93.1034	75	0.5	93.1034	150	0.5	75	8.7	101.8394	Sym 1
98.2759	0	0.5	98.2759	75	0.5	75	8.7	105.9931	Sym 1
98.2759	75	0.5	98.2759	150	0.5	75	8.7	105.046	Sym 1
103.4483	0	0.5	103.4483	75	0.5	75	8.7	108.9136	Sym 1
103.4483	75	0.5	103.4483	150	0.5	75	8.7	107.9662	Sym 1
108.6207	0	0.5	108.6207	75	0.5	75	8.7	113.9608	Sym 1
108.6207	75	0.5	108.6207	150	0.5	75	8.7	113.0138	Sym 1
113.7931	0	0.5	113.7931	75	0.5	75	8.7	120.1239	Sym 1
113.7931	75	0.5	113.7931	150	0.5	75	8.7	119.1871	Sym 1

118.9655	0	0.5	118.9655	75	0.5	75	8.7	126.5495	Sym 1
118.9655	75	0.5	118.9655	150	0.5	75	8.7	125.6268	Sym 1
124.1379	0	0.5	124.1379	75	0.5	75	8.7	136.3114	Sym 1
124.1379	75	0.5	124.1379	150	0.5	75	8.7	135.4307	Sym 1
129.3103	0	0.5	129.3103	75	0.5	75	8.7	147.849	Sym 1
129.3103	75	0.5	129.3103	150	0.5	75	8.7	147.0227	Sym 1
134.4828	0	0.5	134.4828	75	0.5	75	8.7	160.0685	Sym 1
134.4828	75	0.5	134.4828	150	0.5	75	8.7	159.3255	Sym 1
139.6552	0	0.5	139.6552	75	0.5	75	8.7	171.1311	Sym 1
139.6552	75	0.5	139.6552	150	0.5	75	8.7	170.5645	Sym 1
144.8276	0	0.5	144.8276	75	0.5	75	8.7	128.917	Sym 1
144.8276	75	0.5	144.8276	150	0.5	75	8.7	129.1514	Sym 1
150	0	0.5	150	75	0.5	75	8.7	186.4214	Sym 1
150	75	0.5	150	150	0.5	75	8.7	186.3632	Sym 1

Total Length Of Primary Conductors 12424.3 meters

0	0.5	0.5	0	0.5	3.5	3	7.9	6.8745	Asy 1
0.5	0	0.5	0.5	0	3.5	3	7.9	6.9062	Asy 2
1	0	0.5	1	0	3.5	3	7.9	5.5951	Asy 3
1.5	0	0.5	1.5	0	3.5	3	7.9	5.2195	Asy 4
2	0	0.5	2	0	3.5	3	7.9	5.5106	Asy 5
2.5	0	0.5	2.5	0	3.5	3	7.9	7.4166	Asy 6
0.5	0.5	0.5	0.5	0.5	3.5	3	7.9	2.542	Asy 7
0.5	1	0.5	0.5	1	3.5	3	7.9	1.5914	Asy 8
0.5	1.5	0.5	0.5	1.5	3.5	3	7.9	1.4083	Asy 9
0.5	2	0.5	0.5	2	3.5	3	7.9	1.7027	Asy 10
0.5	2.5	0.5	0.5	2.5	3.5	3	7.9	3.355	Asy 11
0	1	0.5	0	1	3.5	3	7.9	5.5384	Asy 12
0	1.5	0.5	0	1.5	3.5	3	7.9	5.1275	Asy 13
0	2	0.5	0	2	3.5	3	7.9	5.3471	Asy 14
0	2.5	0.5	0	2.5	3.5	3	7.9	7.0442	Asy 15
1	0.5	0.5	1	0.5	3.5	3	7.9	1.6088	Asy 16
1	1	0.5	1	1	3.5	3	7.9	0.584	Asy 17
1	1.5	0.5	1	1.5	3.5	3	7.9	0.496	Asy 18
1	2	0.5	1	2	3.5	3	7.9	0.7962	Asy 19
1	2.5	0.5	1	2.5	3.5	3	7.9	2.2236	Asy 20
1.5	0.5	0.5	1.5	0.5	3.5	3	7.9	1.4463	Asy 21
1.5	1	0.5	1.5	1	3.5	3	7.9	0.5121	Asy 22
1.5	1.5	0.5	1.5	1.5	3.5	3	7.9	0.2488	Asy 23
1.5	2	0.5	1.5	2	3.5	3	7.9	0.5678	Asy 24
1.5	2.5	0.5	1.5	2.5	3.5	3	7.9	1.8577	Asy 25
2	0.5	0.5	2	0.5	3.5	3	7.9	1.7881	Asy 26
2	1	0.5	2	1	3.5	3	7.9	0.8465	Asy 27
2	1.5	0.5	2	1.5	3.5	3	7.9	0.5919	Asy 28
2	2	0.5	2	2	3.5	3	7.9	0.6844	Asy 29
2	2.5	0.5	2	2.5	3.5	3	7.9	1.9415	Asy 30
2.5	0.5	0.5	2.5	0.5	3.5	3	7.9	3.6134	Asy 31
2.5	1	0.5	2.5	1	3.5	3	7.9	2.3981	Asy 32
2.5	1.5	0.5	2.5	1.5	3.5	3	7.9	1.9527	Asy 33
2.5	2	0.5	2.5	2	3.5	3	7.9	1.9716	Asy 34
2.5	2.5	0.5	2.5	2.5	3.5	3	7.9	2.9079	Asy 35
0	149.5	0.5	0	149.5	3.5	3	7.9	6.8729	Asy 36
0	149	0.5	0	149	3.5	3	7.9	5.5305	Asy 37
0	148.5	0.5	0	148.5	3.5	3	7.9	5.1128	Asy 38
0	148	0.5	0	148	3.5	3	7.9	5.3223	Asy 39
0	147.5	0.5	0	147.5	3.5	3	7.9	6.9994	Asy 40
0.5	150	0.5	0.5	150	3.5	3	7.9	6.9076	Asy 41
0.5	149.5	0.5	0.5	149.5	3.5	3	7.9	2.5395	Asy 42

0.5	149	0.5	0.5	149	3.5	3	7.9	1.5807	Asy 43
0.5	148.5	0.5	0.5	148.5	3.5	3	7.9	1.3897	Asy 44
0.5	148	0.5	0.5	148	3.5	3	7.9	1.6715	Asy 45
0.5	147.5	0.5	0.5	147.5	3.5	3	7.9	3.2974	Asy 46
1	150	0.5	1	150	3.5	3	7.9	5.5959	Asy 47
1	149.5	0.5	1	149.5	3.5	3	7.9	1.603	Asy 48
1	149	0.5	1	149	3.5	3	7.9	0.5761	Asy 49
1	148.5	0.5	1	148.5	3.5	3	7.9	0.4778	Asy 50
1	148	0.5	1	148	3.5	3	7.9	0.7639	Asy 51
1	147.5	0.5	1	147.5	3.5	3	7.9	2.1626	Asy 52
1.5	150	0.5	1.5	150	3.5	3	7.9	5.2231	Asy 53
1.5	149.5	0.5	1.5	149.5	3.5	3	7.9	1.4414	Asy 54
1.5	149	0.5	1.5	149	3.5	3	7.9	0.5034	Asy 55
1.5	148.5	0.5	1.5	148.5	3.5	3	7.9	0.2364	Asy 56
1.5	148	0.5	1.5	148	3.5	3	7.9	0.54	Asy 57
1.5	147.5	0.5	1.5	147.5	3.5	3	7.9	1.8004	Asy 58
2	150	0.5	2	150	3.5	3	7.9	5.523	Asy 59
2	149.5	0.5	2	149.5	3.5	3	7.9	1.7911	Asy 60
2	149	0.5	2	149	3.5	3	7.9	0.8466	Asy 61
2	148.5	0.5	2	148.5	3.5	3	7.9	0.5888	Asy 62
2	148	0.5	2	148	3.5	3	7.9	0.6746	Asy 63
2	147.5	0.5	2	147.5	3.5	3	7.9	1.9042	Asy 64
2.5	150	0.5	2.5	150	3.5	3	7.9	7.4556	Asy 65
2.5	149.5	0.5	2.5	149.5	3.5	3	7.9	3.6479	Asy 66
2.5	149	0.5	2.5	149	3.5	3	7.9	2.4376	Asy 67
2.5	148.5	0.5	2.5	148.5	3.5	3	7.9	1.9977	Asy 68
2.5	148	0.5	2.5	148	3.5	3	7.9	2.017	Asy 69
2.5	147.5	0.5	2.5	147.5	3.5	3	7.9	2.9398	Asy 70
150	0.5	0.5	150	0.5	3.5	3	7.9	6.8619	Asy 71
150	1	0.5	150	1	3.5	3	7.9	5.5251	Asy 72
150	1.5	0.5	150	1.5	3.5	3	7.9	5.1108	Asy 73
150	2	0.5	150	2	3.5	3	7.9	5.3222	Asy 74
150	2.5	0.5	150	2.5	3.5	3	7.9	6.9945	Asy 75
149.5	0	0.5	149.5	0	3.5	3	7.9	6.894	Asy 76
149.5	0.5	0.5	149.5	0.5	3.5	3	7.9	2.5388	Asy 77
149.5	1	0.5	149.5	1	3.5	3	7.9	1.5865	Asy 78
149.5	1.5	0.5	149.5	1.5	3.5	3	7.9	1.4003	Asy 79
149.5	2	0.5	149.5	2	3.5	3	7.9	1.6863	Asy 80
149.5	2.5	0.5	149.5	2.5	3.5	3	7.9	3.3083	Asy 81
149	0	0.5	149	0	3.5	3	7.9	5.5825	Asy 82
149	0.5	0.5	149	0.5	3.5	3	7.9	1.6043	Asy 83
149	1	0.5	149	1	3.5	3	7.9	0.5824	Asy 84
149	1.5	0.5	149	1.5	3.5	3	7.9	0.4893	Asy 85
149	2	0.5	149	2	3.5	3	7.9	0.7785	Asy 86
149	2.5	0.5	149	2.5	3.5	3	7.9	2.1672	Asy 87
148.5	0	0.5	148.5	0	3.5	3	7.9	5.2041	Asy 88
148.5	0.5	0.5	148.5	0.5	3.5	3	7.9	1.4389	Asy 89
148.5	1	0.5	148.5	1	3.5	3	7.9	0.5059	Asy 90
148.5	1.5	0.5	148.5	1.5	3.5	3	7.9	0.2411	Asy 91
148.5	2	0.5	148.5	2	3.5	3	7.9	0.5444	Asy 92
148.5	2.5	0.5	148.5	2.5	3.5	3	7.9	1.783	Asy 93
148	0	0.5	148	0	3.5	3	7.9	5.4879	Asy 94
148	0.5	0.5	148	0.5	3.5	3	7.9	1.7734	Asy 95
148	1	0.5	148	1	3.5	3	7.9	0.8308	Asy 96
148	1.5	0.5	148	1.5	3.5	3	7.9	0.5702	Asy 97
148	2	0.5	148	2	3.5	3	7.9	0.6459	Asy 98
148	2.5	0.5	148	2.5	3.5	3	7.9	1.8339	Asy 99
147.5	0	0.5	147.5	0	3.5	3	7.9	7.3717	Asy 100
147.5	0.5	0.5	147.5	0.5	3.5	3	7.9	3.5722	Asy 101

147.5	1	0.5	147.5	1	3.5	3	7.9	2.3489	Asy 102
147.5	1.5	0.5	147.5	1.5	3.5	3	7.9	1.8864	Asy 103
147.5	2	0.5	147.5	2	3.5	3	7.9	1.8715	Asy 104
147.5	2.5	0.5	147.5	2.5	3.5	3	7.9	2.721	Asy 105
150	147.5	0.5	150	147.5	3.5	3	7.9	6.9712	Asy 106
150	148	0.5	150	148	3.5	3	7.9	5.3067	Asy 107
150	148.5	0.5	150	148.5	3.5	3	7.9	5.1015	Asy 108
150	149	0.5	150	149	3.5	3	7.9	5.521	Asy 109
150	149.5	0.5	150	149.5	3.5	3	7.9	6.8634	Asy 110
147.5	150	0.5	147.5	150	3.5	3	7.9	7.4427	Asy 111
148	150	0.5	148	150	3.5	3	7.9	5.5151	Asy 112
148.5	150	0.5	148.5	150	3.5	3	7.9	5.2164	Asy 113
149	150	0.5	149	150	3.5	3	7.9	5.5891	Asy 114
149.5	150	0.5	149.5	150	3.5	3	7.9	6.8996	Asy 115
0	0	0.5	0	0	3.5	3	7.9	11.4071	Sym 1
7.8947	0	0.5	7.8947	0	3.5	3	7.9	15.9391	Sym 1
15.7895	0	0.5	15.7895	0	3.5	3	7.9	15.3515	Sym 1
23.6842	0	0.5	23.6842	0	3.5	3	7.9	14.73	Sym 1
31.5789	0	0.5	31.5789	0	3.5	3	7.9	13.6981	Sym 1
39.4737	0	0.5	39.4737	0	3.5	3	7.9	13.3423	Sym 1
47.3684	0	0.5	47.3684	0	3.5	3	7.9	12.795	Sym 1
55.2632	0	0.5	55.2632	0	3.5	3	7.9	12.631	Sym 1
63.1579	0	0.5	63.1579	0	3.5	3	7.9	12.412	Sym 1
71.0526	0	0.5	71.0526	0	3.5	3	7.9	12.3834	Sym 1
78.9474	0	0.5	78.9474	0	3.5	3	7.9	12.4132	Sym 1
86.8421	0	0.5	86.8421	0	3.5	3	7.9	12.4542	Sym 1
94.7368	0	0.5	94.7368	0	3.5	3	7.9	12.6718	Sym 1
102.6316	0	0.5	102.6316	0	3.5	3	7.9	12.8275	Sym 1
110.5263	0	0.5	110.5263	0	3.5	3	7.9	13.359	Sym 1
118.4211	0	0.5	118.4211	0	3.5	3	7.9	13.7035	Sym 1
126.3158	0	0.5	126.3158	0	3.5	3	7.9	14.713	Sym 1
134.2105	0	0.5	134.2105	0	3.5	3	7.9	15.3086	Sym 1
142.1053	0	0.5	142.1053	0	3.5	3	7.9	15.8118	Sym 1
150	0	0.5	150	0	3.5	3	7.9	11.3897	Sym 1
0	150	0.5	0	150	3.5	3	7.9	11.4137	Sym 1
7.8947	150	0.5	7.8947	150	3.5	3	7.9	16.114	Sym 1
15.7895	150	0.5	15.7895	150	3.5	3	7.9	15.5415	Sym 1
23.6842	150	0.5	23.6842	150	3.5	3	7.9	14.8982	Sym 1
31.5789	150	0.5	31.5789	150	3.5	3	7.9	13.8477	Sym 1
39.4737	150	0.5	39.4737	150	3.5	3	7.9	13.4644	Sym 1
47.3684	150	0.5	47.3684	150	3.5	3	7.9	12.8936	Sym 1
55.2632	150	0.5	55.2632	150	3.5	3	7.9	12.7134	Sym 1
63.1579	150	0.5	63.1579	150	3.5	3	7.9	12.4708	Sym 1
71.0526	150	0.5	71.0526	150	3.5	3	7.9	12.4292	Sym 1
78.9474	150	0.5	78.9474	150	3.5	3	7.9	12.4415	Sym 1
86.8421	150	0.5	86.8421	150	3.5	3	7.9	12.4924	Sym 1
94.7368	150	0.5	94.7368	150	3.5	3	7.9	12.7347	Sym 1
102.6316	150	0.5	102.6316	150	3.5	3	7.9	12.9098	Sym 1
110.5263	150	0.5	110.5263	150	3.5	3	7.9	13.4723	Sym 1
118.4211	150	0.5	118.4211	150	3.5	3	7.9	13.8447	Sym 1
126.3158	150	0.5	126.3158	150	3.5	3	7.9	14.8813	Sym 1
134.2105	150	0.5	134.2105	150	3.5	3	7.9	15.5049	Sym 1
142.1053	150	0.5	142.1053	150	3.5	3	7.9	16.0629	Sym 1
150	150	0.5	150	150	3.5	3	7.9	11.4004	Sym 1
0	5	0.5	0	5	3.5	3	7.9	13.6158	Sym 2
150	5	0.5	150	5	3.5	3	7.9	13.4358	Sym 2
0	15	0.5	0	15	3.5	3	7.9	15.5201	Sym 2
150	15	0.5	150	15	3.5	3	7.9	15.4852	Sym 2
0	25	0.5	0	25	3.5	3	7.9	14.3881	Sym 2

150	25	0.5	150	25	3.5	3	7.9	14.3995	Sym 2
0	35	0.5	0	35	3.5	3	7.9	13.5288	Sym 2
150	35	0.5	150	35	3.5	3	7.9	13.5515	Sym 2
0	45	0.5	0	45	3.5	3	7.9	12.9563	Sym 2
150	45	0.5	150	45	3.5	3	7.9	12.9834	Sym 2
0	55	0.5	0	55	3.5	3	7.9	12.6211	Sym 2
150	55	0.5	150	55	3.5	3	7.9	12.6508	Sym 2
0	65	0.5	0	65	3.5	3	7.9	12.429	Sym 2
150	65	0.5	150	65	3.5	3	7.9	12.4612	Sym 2
0	75	0.5	0	75	3.5	3	7.9	12.3253	Sym 2
150	75	0.5	150	75	3.5	3	7.9	12.3662	Sym 2
0	85	0.5	0	85	3.5	3	7.9	12.3747	Sym 2
150	85	0.5	150	85	3.5	3	7.9	12.423	Sym 2
0	95	0.5	0	95	3.5	3	7.9	12.6071	Sym 2
150	95	0.5	150	95	3.5	3	7.9	12.6548	Sym 2
0	105	0.5	0	105	3.5	3	7.9	12.9827	Sym 2
150	105	0.5	150	105	3.5	3	7.9	13.0268	Sym 2
0	115	0.5	0	115	3.5	3	7.9	13.5141	Sym 2
150	115	0.5	150	115	3.5	3	7.9	13.5491	Sym 2
0	125	0.5	0	125	3.5	3	7.9	14.3549	Sym 2
150	125	0.5	150	125	3.5	3	7.9	14.3638	Sym 2
0	135	0.5	0	135	3.5	3	7.9	15.6273	Sym 2
150	135	0.5	150	135	3.5	3	7.9	15.4991	Sym 2
0	145	0.5	0	145	3.5	3	7.9	13.4738	Sym 2
150	145	0.5	150	145	3.5	3	7.9	13.3522	Sym 2
147.5	147.5	0.5	147.5	147.5	3.5	3	7.9	2.9061	Sym 3
148	147.5	0.5	148	147.5	3.5	3	7.9	1.8805	Sym 3
148.5	147.5	0.5	148.5	147.5	3.5	3	7.9	1.7796	Sym 3
149	147.5	0.5	149	147.5	3.5	3	7.9	2.1428	Sym 3
149.5	147.5	0.5	149.5	147.5	3.5	3	7.9	3.2765	Sym 3
147.5	148	0.5	147.5	148	3.5	3	7.9	2.0001	Sym 3
148	148	0.5	148	148	3.5	3	7.9	0.6674	Sym 3
148.5	148	0.5	148.5	148	3.5	3	7.9	0.5341	Sym 3
149	148	0.5	149	148	3.5	3	7.9	0.7578	Sym 3
149.5	148	0.5	149.5	148	3.5	3	7.9	1.6636	Sym 3
147.5	148.5	0.5	147.5	148.5	3.5	3	7.9	1.9866	Sym 3
148	148.5	0.5	148	148.5	3.5	3	7.9	0.5852	Sym 3
148.5	148.5	0.5	148.5	148.5	3.5	3	7.9	0.235	Sym 3
149	148.5	0.5	149	148.5	3.5	3	7.9	0.4755	Sym 3
149.5	148.5	0.5	149.5	148.5	3.5	3	7.9	1.3854	Sym 3
147.5	149	0.5	147.5	149	3.5	3	7.9	2.4288	Sym 3
148	149	0.5	148	149	3.5	3	7.9	0.844	Sym 3
148.5	149	0.5	148.5	149	3.5	3	7.9	0.5021	Sym 3
149	149	0.5	149	149	3.5	3	7.9	0.5752	Sym 3
149.5	149	0.5	149.5	149	3.5	3	7.9	1.5776	Sym 3
147.5	149.5	0.5	147.5	149.5	3.5	3	7.9	3.6393	Sym 3
148	149.5	0.5	148	149.5	3.5	3	7.9	1.7878	Sym 3
148.5	149.5	0.5	148.5	149.5	3.5	3	7.9	1.4391	Sym 3
149	149.5	0.5	149	149.5	3.5	3	7.9	1.6008	Sym 3
149.5	149.5	0.5	149.5	149.5	3.5	3	7.9	2.5365	Sym 3
10	5	0.5	10	5	3.5	3	7.9	10.6467	Sym 14
15.625	5	0.5	15.625	5	3.5	3	7.9	10.4279	Sym 14
21.25	5	0.5	21.25	5	3.5	3	7.9	10.3193	Sym 14
26.875	5	0.5	26.875	5	3.5	3	7.9	10.1573	Sym 14
32.5	5	0.5	32.5	5	3.5	3	7.9	9.9522	Sym 14
38.125	5	0.5	38.125	5	3.5	3	7.9	9.7813	Sym 14
43.75	5	0.5	43.75	5	3.5	3	7.9	9.6569	Sym 14
49.375	5	0.5	49.375	5	3.5	3	7.9	9.4724	Sym 14
55	5	0.5	55	5	3.5	3	7.9	9.2863	Sym 14

60.625	5	0.5	60.625	5	3.5	3	7.9	9.1654	Sym 14
66.25	5	0.5	66.25	5	3.5	3	7.9	9.0125	Sym 14
71.875	5	0.5	71.875	5	3.5	3	7.9	8.8337	Sym 14
77.5	5	0.5	77.5	5	3.5	3	7.9	8.6909	Sym 14
83.125	5	0.5	83.125	5	3.5	3	7.9	8.8787	Sym 14
88.75	5	0.5	88.75	5	3.5	3	7.9	9.0542	Sym 14
94.375	5	0.5	94.375	5	3.5	3	7.9	9.2235	Sym 14
100	5	0.5	100	5	3.5	3	7.9	9.4601	Sym 14
105.625	5	0.5	105.625	5	3.5	3	7.9	9.6771	Sym 14
111.25	5	0.5	111.25	5	3.5	3	7.9	9.8477	Sym 14
116.875	5	0.5	116.875	5	3.5	3	7.9	10.0883	Sym 14
122.5	5	0.5	122.5	5	3.5	3	7.9	10.3468	Sym 14
128.125	5	0.5	128.125	5	3.5	3	7.9	10.5445	Sym 14
133.75	5	0.5	133.75	5	3.5	3	7.9	10.7194	Sym 14
139.375	5	0.5	139.375	5	3.5	3	7.9	10.536	Sym 14
145	5	0.5	145	5	3.5	3	7.9	8.2272	Sym 14
10	140	0.5	10	140	3.5	3	7.9	9.7778	Sym 14
15.625	140	0.5	15.625	140	3.5	3	7.9	9.7475	Sym 14
21.25	140	0.5	21.25	140	3.5	3	7.9	9.746	Sym 14
26.875	140	0.5	26.875	140	3.5	3	7.9	9.6545	Sym 14
32.5	140	0.5	32.5	140	3.5	3	7.9	9.5371	Sym 14
38.125	140	0.5	38.125	140	3.5	3	7.9	9.4011	Sym 14
43.75	140	0.5	43.75	140	3.5	3	7.9	9.2743	Sym 14
49.375	140	0.5	49.375	140	3.5	3	7.9	9.137	Sym 14
55	140	0.5	55	140	3.5	3	7.9	8.9967	Sym 14
60.625	140	0.5	60.625	140	3.5	3	7.9	8.8597	Sym 14
66.25	140	0.5	66.25	140	3.5	3	7.9	8.7113	Sym 14
71.875	140	0.5	71.875	140	3.5	3	7.9	8.5749	Sym 14
77.5	140	0.5	77.5	140	3.5	3	7.9	8.4248	Sym 14
83.125	140	0.5	83.125	140	3.5	3	7.9	8.5759	Sym 14
88.75	140	0.5	88.75	140	3.5	3	7.9	8.7616	Sym 14
94.375	140	0.5	94.375	140	3.5	3	7.9	8.9375	Sym 14
100	140	0.5	100	140	3.5	3	7.9	9.1183	Sym 14
105.625	140	0.5	105.625	140	3.5	3	7.9	9.304	Sym 14
111.25	140	0.5	111.25	140	3.5	3	7.9	9.4778	Sym 14
116.875	140	0.5	116.875	140	3.5	3	7.9	9.6653	Sym 14
122.5	140	0.5	122.5	140	3.5	3	7.9	9.8407	Sym 14
128.125	140	0.5	128.125	140	3.5	3	7.9	9.9876	Sym 14
133.75	140	0.5	133.75	140	3.5	3	7.9	10.0463	Sym 14
139.375	140	0.5	139.375	140	3.5	3	7.9	9.6145	Sym 14
145	140	0.5	145	140	3.5	3	7.9	9.2637	Sym 14
5	3	0.5	5	3	3.5	3	7.9	8.614	Sym 15
145	3	0.5	145	3	3.5	3	7.9	7.8801	Sym 15
5	8.9167	0.5	5	8.9167	3.5	3	7.9	11.2958	Sym 15
145	8.9167	0.5	145	8.9167	3.5	3	7.9	10.9823	Sym 15
5	14.8333	0.5	5	14.8333	3.5	3	7.9	10.8765	Sym 15
145	14.8333	0.5	145	14.8333	3.5	3	7.9	10.8592	Sym 15
5	20.75	0.5	5	20.75	3.5	3	7.9	10.2037	Sym 15
145	20.75	0.5	145	20.75	3.5	3	7.9	10.231	Sym 15
5	26.6667	0.5	5	26.6667	3.5	3	7.9	10.1423	Sym 15
145	26.6667	0.5	145	26.6667	3.5	3	7.9	10.1835	Sym 15
5	32.5833	0.5	5	32.5833	3.5	3	7.9	10.0635	Sym 15
145	32.5833	0.5	145	32.5833	3.5	3	7.9	10.1108	Sym 15
5	38.5	0.5	5	38.5	3.5	3	7.9	9.9291	Sym 15
145	38.5	0.5	145	38.5	3.5	3	7.9	9.9796	Sym 15
5	44.4167	0.5	5	44.4167	3.5	3	7.9	9.619	Sym 15
145	44.4167	0.5	145	44.4167	3.5	3	7.9	9.6712	Sym 15
5	50.3333	0.5	5	50.3333	3.5	3	7.9	9.4295	Sym 15
145	50.3333	0.5	145	50.3333	3.5	3	7.9	9.4832	Sym 15

5	56.25	0.5	5	56.25	3.5	3	7.9	9.022	Sym 15
145	56.25	0.5	145	56.25	3.5	3	7.9	9.0765	Sym 15
5	62.1667	0.5	5	62.1667	3.5	3	7.9	8.7543	Sym 15
145	62.1667	0.5	145	62.1667	3.5	3	7.9	8.8101	Sym 15
5	68.0833	0.5	5	68.0833	3.5	3	7.9	8.9965	Sym 15
145	68.0833	0.5	145	68.0833	3.5	3	7.9	9.0548	Sym 15
5	74	0.5	5	74	3.5	3	7.9	9.0802	Sym 15
145	74	0.5	145	74	3.5	3	7.9	9.1441	Sym 15
5	79.9167	0.5	5	79.9167	3.5	3	7.9	9.2487	Sym 15
145	79.9167	0.5	145	79.9167	3.5	3	7.9	9.3208	Sym 15
5	85.8333	0.5	5	85.8333	3.5	3	7.9	9.1818	Sym 15
145	85.8333	0.5	145	85.8333	3.5	3	7.9	9.255	Sym 15
5	91.75	0.5	5	91.75	3.5	3	7.9	9.2071	Sym 15
145	91.75	0.5	145	91.75	3.5	3	7.9	9.2803	Sym 15
5	97.6667	0.5	5	97.6667	3.5	3	7.9	9.0878	Sym 15
145	97.6667	0.5	145	97.6667	3.5	3	7.9	9.1597	Sym 15
5	103.5833	0.5	5	103.5833	3.5	3	7.9	9.02	Sym 15
145	103.5833	0.5	145	103.5833	3.5	3	7.9	9.0897	Sym 15
5	109.5	0.5	5	109.5	3.5	3	7.9	9.5613	Sym 15
145	109.5	0.5	145	109.5	3.5	3	7.9	9.628	Sym 15
5	115.4167	0.5	5	115.4167	3.5	3	7.9	9.9189	Sym 15
145	115.4167	0.5	145	115.4167	3.5	3	7.9	9.9794	Sym 15
5	121.3333	0.5	5	121.3333	3.5	3	7.9	10.4311	Sym 15
145	121.3333	0.5	145	121.3333	3.5	3	7.9	10.4799	Sym 15
5	127.25	0.5	5	127.25	3.5	3	7.9	10.7758	Sym 15
145	127.25	0.5	145	127.25	3.5	3	7.9	10.7951	Sym 15
5	133.1667	0.5	5	133.1667	3.5	3	7.9	11.0815	Sym 15
145	133.1667	0.5	145	133.1667	3.5	3	7.9	10.9891	Sym 15
5	139.0833	0.5	5	139.0833	3.5	3	7.9	11.27	Sym 15
145	139.0833	0.5	145	139.0833	3.5	3	7.9	9.4573	Sym 15
5	145	0.5	5	145	3.5	3	7.9	8.9602	Sym 15
145	145	0.5	145	145	3.5	3	7.9	8.7439	Sym 15

Total Length Of Primary Rods 930 meters

Total Number Of Elements 474

Ground Potential Rise 5732.27 volts
Calculated Ground Resistance 0.268722 ohms
Equivalent Impedance 0.188105 ohms